SYNCLAB

Centralized Workshop & Laboratory Allocation Platform

Group 01
NANO 2142 Introduction to Software Development

Faculty of Technology
Wayamba University of Sri Lanka

Prepared 07 Oct 2026 as supporting project overview
Prior initial submission or presentation not verified

Problem and objectives

Students need a clear way to request laboratory sessions and track administrator decisions.

Availability

Show free dates and time slots for the selected laboratory.

Request details

Collect student information, supervisor and project purpose.

Administrator review

Record approval, rejection and remarks for each allocation.

Persistent records

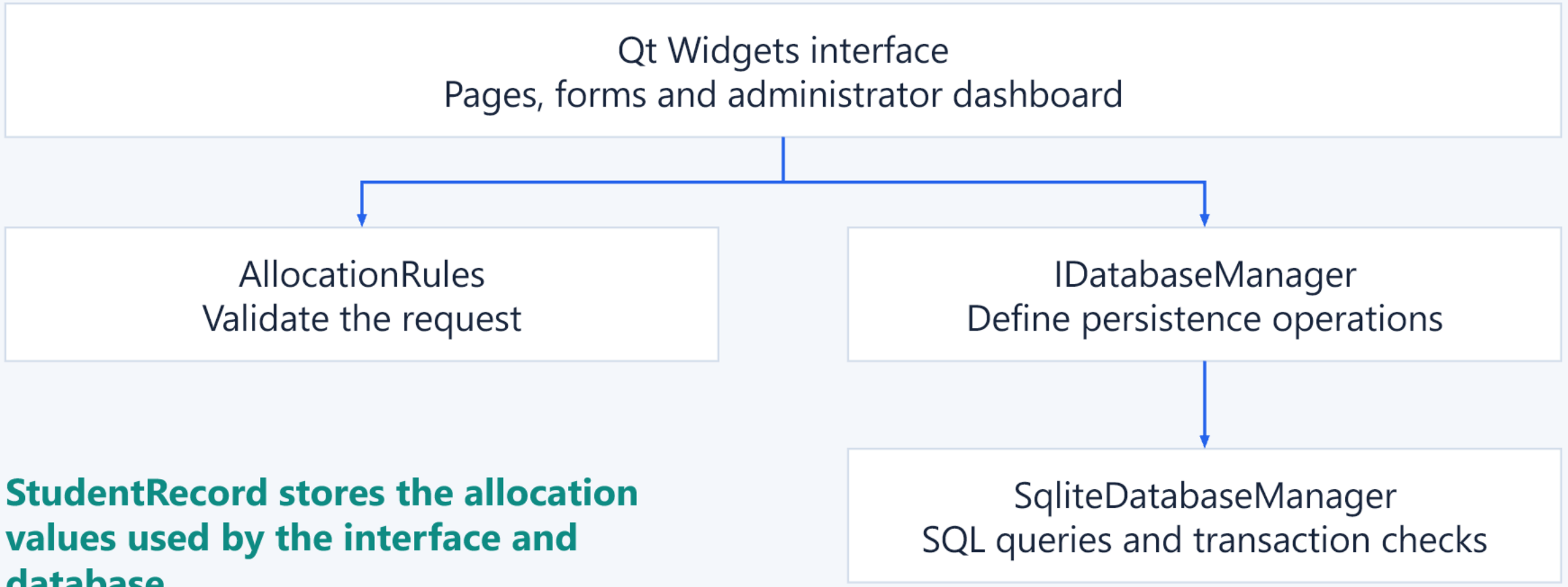
Keep allocation data available for updates and cancellation.

System scope and course components

Component	Project scope
Qt interface	Five pages for selection, date/time, student form and review
SQL database	Persist students, workshops, allocations and individual slots
C++ OOP	StudentRecord encapsulation and an abstract database interface
Allocation logic	Ten laboratory choices, four daily slots and conflict checks
Integration	Connect UI events to validation and database operations

Desktop application with local SQLite storage

System architecture



StudentRecord stores the allocation values used by the interface and database.

Data and workshop allocation flow

Relation	Stored fact
students	Student profile
workshops	Category and department
records	Allocation and project details
booking_slots	Selected time slots

Each allocation refers to one student and one laboratory.

1. Select a laboratory
2. Choose date and free slots
3. Enter student and project details
4. Validate and save as Pending
5. Administrator makes a decision

Proposed roles and five-minute plan

Member names and role assignments require group confirmation.

Recommended role	Suggested speaking topic	Time
UI/UX Designer	Slides 1-2: idea, problem and objectives	1:00
OOP Developer	Slide 3: system scope and core logic	1:00
Integrator	Slide 4: component architecture	1:00
Database Designer	Slide 5: data and allocation flow	1:00
Documentation & Testing Lead	Slide 6: roles, deliverables and conclusion	1:00

Total planned duration: 5:00 minutes